

CHEATSHEET · QUICK REFERENCE

Lab 11 — Directory Services with OpenLDAP

Build an OpenLDAP tree with OUs, users and groups, delegate administration by group via an olcAccess ACL, and relate it to Active Directory.

Verify and structure the directory

```
ldapsearch -x -H ldap://127.0.0.1 -b "dc=lab,dc=local" -s base namingContexts
Confirm the base DN

ldapadd -x -D "cn=admin,dc=lab,dc=local" -w LabAdmin1! -f /tmp/ou.ldif
Add OUs (people, groups)

ldapadd -x -D "cn=admin,dc=lab,dc=local" -w LabAdmin1! -f /tmp/users.ldif
Add user entries

ldappasswd -x -D "cn=admin,dc=lab,dc=local" -w LabAdmin1! -s 'AlicePass!' "uid=alice,ou=people,dc=lab,dc=local"
Set a user password
```

Bind, search and delegate

```
ldapwhoami -x -D "uid=alice,ou=people,dc=lab,dc=local" -w 'AlicePass!'
Bind as a user

ldapsearch -x -D "uid=alice,..." -w 'AlicePass!' -b "ou=people,dc=lab,dc=local" "(objectclass=inetorgperson)"
Search entries

ldapmodify -Y EXTERNAL -H ldapi:/// -f /tmp/acl.ldif
Apply an olcAccess delegation ACL
```

Concept mapping

LDAP concept	Maps to
Base DN -> OU -> entry	Directory tree structure
organizationalUnit	Scope
groupOfNames	Role
olcAccess ACL	Delegation
sssd / nslcd on client	How Linux consumes the directory
Active Directory	LDAP + Kerberos, same concepts